

system, e.g. increased efficiency, better reliability, and better responsiveness towards societal challenges; (ii) on the innovation system, e.g. faster innovation when results are shared earlier, and innovations more directed towards societal challenges; (iii) on the interface between science and society, e.g. more productive interactions among academia and other knowledge actors, and higher trust of society in the science system when researchers and citizens are engaged. While past projects have started building an evidence base, this remains fragmented and incomplete. A broad and comprehensive evidence base would help define new policies for open science, drive further uptake and help communicate on open science.

Proposals are expected to:

- systematise and evaluate the validity and robustness of existing literature, data and evidence of impacts of open science practice, including potential legal and licensing issues;
- leverage and valorise the body of knowledge resulting from the Science and Society (FP6), Science in Society (FP7) and Science with and for Society (Horizon 2020) programmes;
- complement existing evidence and develop scientific methodologies and models to capture impacts, notably those relating to socio-economic, including gender equality related, environmental and public health aspects. It is in particular expected to develop and implement methods for measuring the contribution of open science practice to the reproducibility of research results, and the implications of involving citizens, civil society and end-users in R&I;
- perform cost/benefit analyses of open science practice and conduct research to identify by which causality/mechanisms the impacts develop.

HORIZON-WIDERA-2021-ERA-01-41: Global cooperation on FAIR data policy and practice

Specific conditions	
<i>Expected EU contribution per project</i>	The Commission estimates that an EU contribution of around EUR 1.00 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.
<i>Indicative budget</i>	The total indicative budget for the topic is EUR 2.00 million.
<i>Type of Action</i>	Coordination and Support Actions
<i>Eligibility conditions</i>	The conditions are described in General Annex B. The following exceptions apply: Legal entities established in non-associated third countries may

	exceptionally participate in this Coordination and support action.
<i>Legal and financial set-up of the Grant Agreements</i>	The rules are described in General Annex G. The following exceptions apply: Beneficiaries will be subject to the following additional dissemination obligations: Beneficiaries must make proactive efforts to freely share, in a timely manner and as appropriate, all relevant results with the other grants awarded from the same call topic, and with the EOSC Partnership⁴⁴. Beneficiaries must acknowledge and incorporate these obligations in the proposal, outlining the efforts they will make towards meeting them, and in Annex I to the Grant Agreement. Beneficiaries will be subject to the additional exploitation obligations: Beneficiaries must make all relevant results generated in the action available for re-use, through a well-defined mechanism, to the EOSC Partnership. Beneficiaries must acknowledge and incorporate these obligations in the proposal, outlining the efforts they will make towards meeting them and in Annex I to the Grant Agreement.

Expected Outcome: Projects are expected to contribute to the following expected outcomes:

- Strengthened international cooperation to increase and mainstream FAIRness⁴⁵ of data and digital objects.
- Connection of disconnected initiatives on data management, data stewardship and FAIR data practices, across borders and disciplines, as enablers of open science.
- Increased FAIR data sharing within and across scientific disciplines and innovation sectors.

These targeted outcomes in turn contribute to medium and long-term impacts:

- Proliferation of interdisciplinary research that helps address societal challenges.
- More efficient research practices as a result of an increased reproducibility of research and reduced duplication of efforts.
- Better informed citizens and society about the results and value of research.
- Improved quality of R&I within the EU.
- Contributions to sustainable growth and faster innovation in Europe, and beyond, in the context of the global economy.

⁴⁴ https://ec.europa.eu/info/files/european-open-science-cloud-eosc_en

⁴⁵ 'FAIRness' is the compliance with the requirements of FAIR data.

Scope: Technological advancements have made science more data intensive and interconnected, with researchers producing and sharing increasing volumes of research data. To produce high quality research data, researchers have to follow good data management and data stewardship practices. Beyond proper data collection, annotation and archival, good data management and stewardship include long-term care of valuable digital assets, either alone or in combination with newly generated data. To maximise the value of science, research data should be FAIR: Findable, Accessible, Interoperable and Reusable. The FAIR principles, introduced in 2014, are a minimal set of community-agreed guiding principles that allow both machines and humans to find, access, interoperate and re-use research data. It is recognised that FAIR data play an essential role in the objectives of Open Science to improve and accelerate scientific research, to increase the engagement of society, and to contribute significantly to economic growth. Accordingly, the EU's Open Science policy⁴⁶ contains the ambition to make FAIR data sharing the default for scientific research and this can be accelerated by focusing on specific scientific disciplines. Although the FAIR principles were initially applied to research data, their coverage extends to all digital objects that are essential to research practice (e.g. algorithms, models, tools, workflows), and to other public sector data. However, initiatives for good data management and stewardship practices and FAIR practice remain fragmented across borders and disciplines. In addition, interoperability remains the least developed to date. Interoperability standards, at discipline-level first, and then across disciplines, are an essential catalyst to foster interdisciplinary science to tackle the global societal challenges of our age. Finally, FAIR digital objects related to the research process are increasingly indispensable to ensure the reproducibility, integrity and re-use of data.

Proposals should support international cooperation on the FAIRness of both data & digital objects in a discipline-specific manner. Applicants should map current initiatives and best practices, globally, within a given scientific discipline, and should facilitate the exchange of best practices across disciplines. They should support case studies and pilots to implement both domain-specific and domain-independent recommendations in FAIR practice (from the Research Data Alliance –RDA-, the Committee on Data of the International Science Council - CODATA-, etc.). They should develop, pilot and possibly deploy interoperability standards and guidelines for increasing FAIRness in specific scientific disciplines, and across different disciplines. They should also develop assessment and evaluation methodologies to appraise FAIRness within disciplines and to develop domain-specific benchmarks.

To ensure complementarity of outcomes, proposals are expected to cooperate and align with activities of the European Open Science Cloud (EOSC) Partnership and to coordinate with relevant initiatives and projects contributing to the development of EOSC. In particular, in areas such as data interoperability, metadata and vocabularies or the use of persistent identifiers, proposals should coordinate the work and establish a feedback mechanism with the awarded proposal(s) from the topic HORIZON-INFRA-2021-EOSC-01-05 in order to ensure alignment with EOSC policies and to identify common useful tools and resources as well as relevant data repositories that comply with EOSC guidelines.

⁴⁶ <https://ec.europa.eu/digital-single-market/en/open-science>